3 An experiment is carried out, using a light intensity meter attached to a toy train, to measure the diffraction of laser light through a double slit, as shown in Fig. 3.1 below.

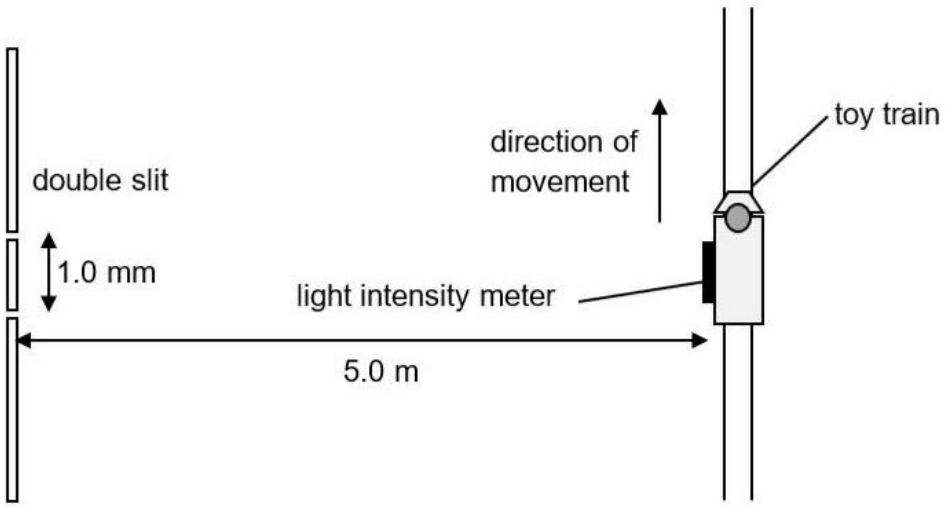


Fig. 3.1

Light from the laser is incident on the double slit. The light from the double slit is then detected by a light sensor attached to the train.

As the train moves along the track, the light intensity is recorded by the light intensity meter. The variation with time of the light intensity is shown in Fig. 3.2.

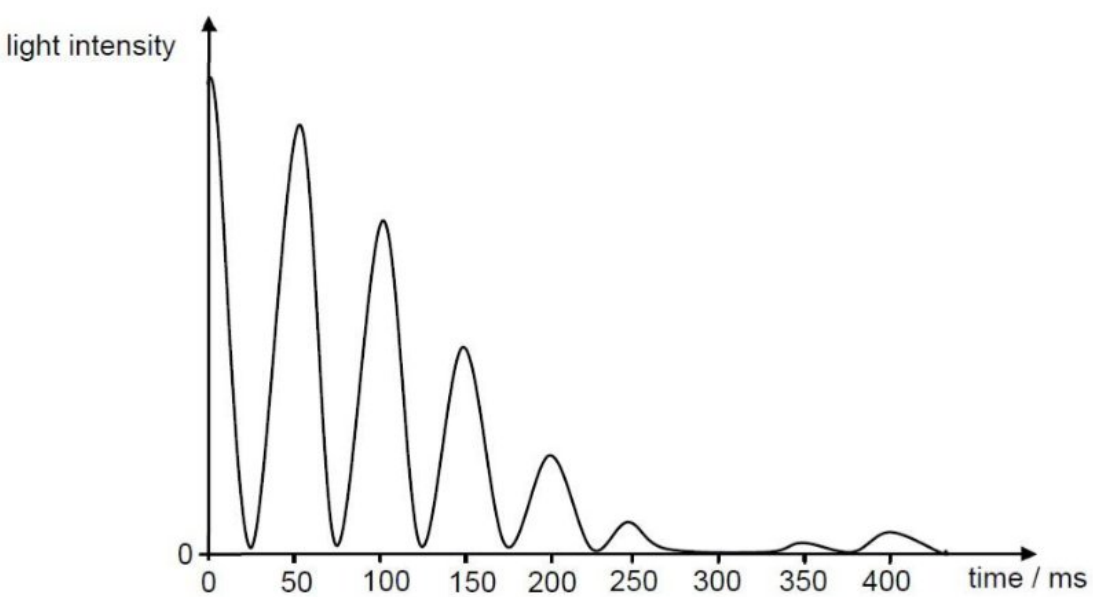


Fig. 3.2

(a) Explain, with reference to the paths travelled by the light emerging from each slit, why maxima and minima are observed.

.....

.....

.....

.....

.....

.....

..... [2]

(b) **(i)** The speed of the train is 6.5 cm s^{-1}.

Use Fig. 3.2 to show that the fringe separation of the two slit interference pattern is 3.3×10^{-3} m.

[1]

(ii) The separation of the slits is 1.0 mm and the track is 5.0 m from the slits.

Determine the wavelength of the laser light.

wavelength = ... m [2]

(c) A maximum is detected by the light intensity meter at every 50 ms, explain why a minimum is detected at 300 ms in Fig. 3.2.

.....

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.....

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.....

.....

..... [1]

(d) Determine the width of one of the slits.

width = ... m [2]

(e) The experiment is changed so that the toy car is now used to investigate the formation of a stationary sound wave as shown in Fig. 3.3 below.

The light sensor of the train is replaced with a sound sensor attached to a data logger. A loudspeaker emitting sound waves of constant amplitude and frequency is placed such that it faces a reflecting barrier. The train track is adjusted so that the train is moving away from the loudspeaker towards the barrier.

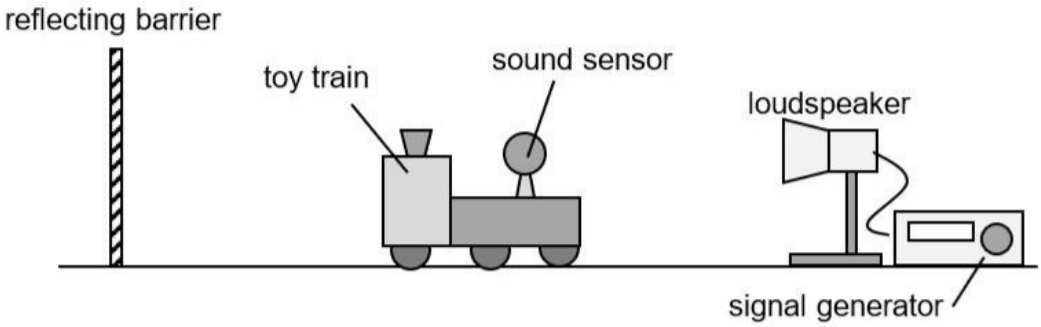


Fig. 3.3

The variation with distance of the sound intensity recorded by the sound sensor is shown in Fig. 3.4.

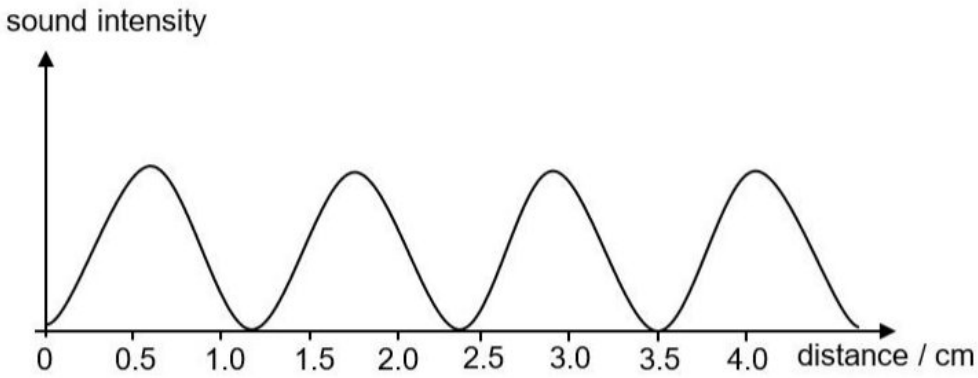


Fig. 3.4

(i) Explain how a stationary sound wave forms between the loudspeaker and reflecting barrier.

.....

.....

.....

.....

.....

..... [2]

(ii) Using Fig. 3.4, determine the wavelength of the stationary wave.

wavelength = ... m [2]

(iii) Given that the frequency of the sound is 14 kHz.
Determine the speed of the sound wave.

speed = ... m s^{-1} [2]

[Total: 14]